

KAĞAN ONUR

Data Analyst • Data Scientist

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PROFILE

Statistics graduate (Yıldız Technical University, 2026) who builds complete analyses rather than isolated models. Modelled 252 months of Turkish macro-financial data with ARIMA, GARCH/EGARCH and LSTM; estimated fixed-effects panel models on 108 observations with cluster-robust inference; represented Türkiye at ASA Data Quest 2025 against teams from six continents. Comfortable across Python, R, Stata and SQL, and equally focused on explaining *why* a model behaves the way it does. Looking for a data analyst or data scientist role where statistical rigour and clear communication both matter.

SELECTED PROJECTS

Time Series Analysis of Turkish Investment Instruments

Undergraduate Thesis • Yıldız Technical University • 2026

Advisor: Prof. Dr. Öyküm Esra Yiğit • Python (statsmodels, arch, TensorFlow/Keras)

- Modelled **252 monthly observations (Jan 2005 – Dec 2025)** of spot gold, USD/TRY, CPI inflation, deposit rates and the BIST 100 within a single econometric framework.
- Ran ADF / PP / KPSS unit-root tests and **Bai-Perron** endogenous break detection, which identified the 2008, 2018 and 2021 breaks; adding break dummies lifted the gold model from $R^2 = 0.12$ to **0.51**.
- Tested long-run equilibrium with **ARDL bounds testing**; finding no definitive cointegration, switched deliberately to independent ARIMA/SARIMA specifications rather than forcing a VECM.
- Estimated **ARMA-GARCH(1,1)** and **EGARCH** volatility models — persistence $\alpha + \beta \approx 0.97$ – 0.98 , with asymmetric leverage effects confirmed across all series.
- Benchmarked ARIMA against a **2×64-neuron LSTM** (lookback 24, dropout 0.20, Adam) and Holt on a 46-month hold-out; ARIMA reached $R^2 = 0.988$ (USD/TRY), 0.979 (gold), 0.975 (deposit rate) and 0.961 (BIST 100).
- Diagnosed *why* the parsimonious model won — rolling one-step-ahead forecasting versus the LSTM's autoregressive error accumulation — instead of reporting the metric alone.
- Granger causality mapped the transmission channels: BIST 100 ($p < 0.01$) and deposit rates ($p < 0.05$) drive gold, while USD/TRY drives inflation ($p < 0.001$), quantifying exchange-rate pass-through.

Effect of Price Discounts on Player Counts on Steam

Static Panel Data Analysis • Team of 3 • 2026

Stata, R • balanced panel of 9 games × 12 months (108 observations)

- Engineered a **weighted discount intensity** variable (discount rate × discounted days ÷ 30) that collapsed two collinear regressors into one interpretable measure of monthly discount pressure.
- Verified stationarity with **Levin-Lin-Chu**, **Im-Pesaran-Shin** and **Fisher-ADF** panel unit-root tests; differenced non-stationary $\ln(\text{price})$ before estimation.
- Used the **Hausman test** ($\chi^2 = 9.27$, $p = 0.026$) to select fixed effects over random effects, then re-estimated with game-clustered robust standard errors.
- Key result: a **1% price increase reduces player count by 0.57%** ($\beta = -0.567$, $p = 0.002$), while seasonal sale dummies showed no significant within-game effect.
- Separated between-game from within-game variation, showing that game size alone explains 98% of the cross-sectional spread — a distinction that changes the marketing conclusion entirely.

Data Science & Sustainable Development

ASA Data Quest 2025 • International • Representing Türkiye

American Statistical Association (CIRS) • Mentor: Ceylan Yozgatlıgil • Team of 4

- Presented Türkiye's entry alongside teams from Brazil, Indonesia, India, Kenya and China in an international student data science competition.
- Built a **four-dimensional sustainability optimisation model** combining economic capacity, education, health expenditure and inverted CO₂ emissions; Nordic countries and Costa Rica emerged with the most balanced profiles.
- Quantified the development linkage: GDP–education correlation of **0.87–0.90** and education–life expectancy of **0.92**.
- Showed that GDP correlates 0.87 with CO₂ while education and health indicators correlate weakly or negatively — evidence that social investment is the less carbon-intensive growth path.

- Validated the specification with Durbin–Watson, VIF and Shapiro–Wilk diagnostics.

Customer Purchasing Behaviour Analysis

Team of 3 • 2026

- Applied multivariate logistic regression and Pearson chi-square tests to identify the drivers of purchase decisions (age, gender, review quality); gender emerged as a statistically significant predictor.

EXPERIENCE

Data Annotation Specialist Jan 2026 – Feb 2026

Artiwise — AI-Powered Software Company

- Labelled and classified text data for supervised NLP models at a company building AI-powered document analysis for customer-centric process management.
- Applied and refined annotation guidelines to keep labels consistent, and ran quality checks on completed batches to keep training data reliable.

EDUCATION

Yıldız Technical University 2022 – 2026

BSc Statistics, Faculty of Arts and Sciences

Relevant coursework: Econometrics, Time Series Analysis, Panel Data Analysis, Multivariate Statistics, Regression Analysis, Data Visualization, Data Management

İstanbul Kültür University 2020 – 2022

Computer Engineering, Faculty of Engineering

English preparatory programme completed; transferred to Statistics at YTU

Gürlek Nakipoğlu Anatolian High School 2015 – 2019

TECHNICAL SKILLS

Programming & Tools	Python (pandas, NumPy, statsmodels, arch, scikit-learn, TensorFlow/Keras) • R • Stata • SQL • Excel / Google Sheets • SPSS
Time Series & Econometrics	ARIMA / SARIMA • ARMA-GARCH • EGARCH • ARDL bounds testing • Granger causality • Bai–Perron structural breaks • ADF / PP / KPSS • STL decomposition
Panel & Regression	Fixed and random effects • Hausman test • cluster-robust standard errors • panel unit-root tests • OLS • logistic and multivariate regression • hypothesis testing
Machine Learning	LSTM and deep sequence models • supervised classification • train/validation/test design • model benchmarking (MAE, MAPE, R ²)
Data Work	Cleaning and merging multi-source data • feature engineering • data labelling and structuring for supervised learning • statistical visualization and reporting

LANGUAGES & CERTIFICATIONS

Languages	Turkish — Native • English — Upper-Intermediate (B2)
Certifications	Upper-Intermediate English Course — DİLKO, in partnership with Pearson (2022)